Directorate of Artificial Intelligence & Visualisation,
Research Centre Imarat (RCI), Defence Research & Development Organisation (DRDO)
&
Department of Computer Science,
Defence Institute of Advanced Technology (DIAT)

LITERATURE SURVEY

ON

CROSS-SPECTRAL DSMAC

Presented by:
Mohineet Kaur (25-26-08)
M. Tech (Artificial Intelligence)

Guide:
Prof. CRS Kumar
A. Kalidasu (Scientist-F)

INTRODUCTION

What is DSMAC?

- DSMAC (Digital Scene Matching Area Correlation) is a vision-based navigation technique that estimates a vehicle's position by matching onboard sensor imagery with reference maps.
- It is widely used in GPS-denied environments where satellite navigation is unavailable or unreliable.
- DSMAC compares observed scene features with stored terrain or scene information to determine accurate localization.

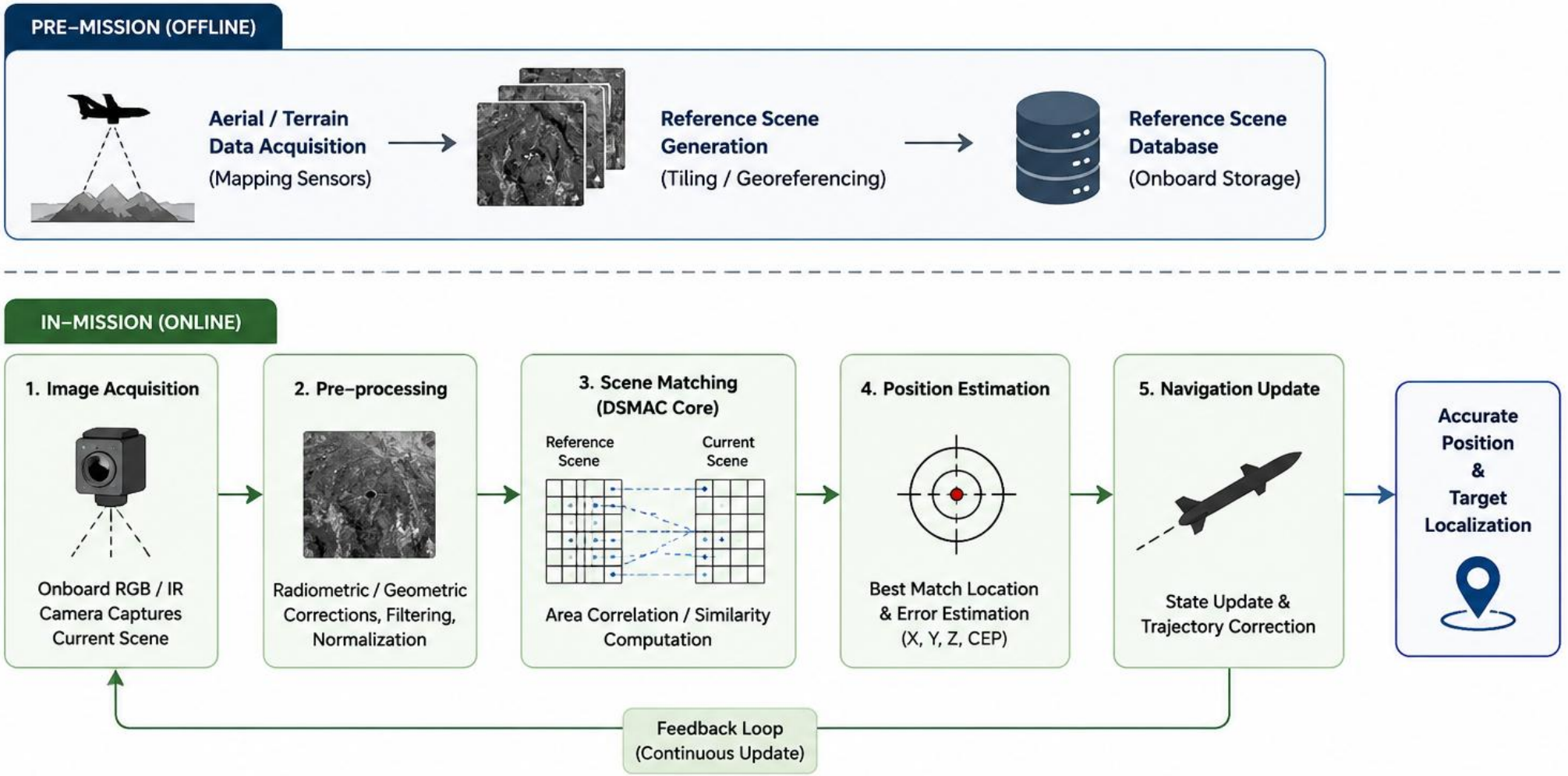
Why is DSMAC Important?

- Provides autonomous navigation without dependence on GPS.
- Enhances robustness in contested or signal-denied environments.
- Applicable to aerial vehicles, missiles, autonomous robots, and unmanned systems.
- Enables precise localization using onboard vision sensors.

Motivation

Modern navigation systems increasingly employ multiple sensing modalities, such as RGB and SAR (Synthetic Aperture Radar) seeker, to improve robustness under varying environmental and illumination conditions.

DSMAC – GENERAL WORKING PIPELINE [1]



EVOLUTION OF DSMAC

1980

Original DSMAC [1]

Approach

- Area-based image correlation
- Correlation Surface Peak Detection
- Pixel-wise scene comparison
- DSMAC-assisted terminal guidance
- Reference image matching

Limitations

- Sensitive to illumination and viewpoint changes
- High computational complexity
- Limited adaptability

1994

Advanced Scene Matching [2]

Approach

- Enhanced image processing techniques
- Improved scene registration
- Robust geometric alignment
- Optimized image matching for terminal guidance

Limitations

- Still relied on handcrafted image processing
- No learned feature representation
- Sensitive to environmental variations

2026

AI-Based DSMAC [3]

Approach

- Deep feature representation
- ResNet-50 feature learning
- FAISS similarity search
- CNN-based aerial image matching
- Automated dataset generation

Limitations

- High computational requirements
- Limited multimodal (RGB–IR) capability

Proposed Work

Lightweight Cross-Modal (RGB–SAR) Scene Matching for Edge Deployment

LITERATURE REVIEW

Method / Paper	Year	Methodology / Pipeline	Contribution	Limitations
LoFTR [4]	2021	CNN features → Transformer self/cross-attention → coarse matching → fine refinement	Detector-free dense correspondence, effective in low-texture regions.	Computationally intensive; not designed specifically for SAR–optical matching.
EfficientLoFTR [5]	2024	CNN → adaptive token selection → efficient Transformer → coarse-to-fine matching	mproves LoFTR efficiency while retaining strong matching performance.	Still not SAR–RGB specific; Transformer computation remains.
XoFTR [6]	2024	Multi-scale CNN → coarse matching → fine matching → sub-pixel refinement	Extends detector-free matching to visible–thermal cross-modal imagery.	Demonstrated on visible–thermal, not SAR–RGB; no DSMAC localization.
Chen et al. [7]	2023	DCNN → dense features → CAMM → keypoints/descriptors → matching	Pair-based SAR–optical registration with modality-aware co-attention.	Keypoint-based; training-dependent; not a complete DSMAC pipeline.
MatchAnything [8]	2025	SAM → object/region proposals → feature learning → cross-domain association	Generalizable cross-domain visual association using segmentation foundation models.	Designed for object association/tracking, not geometric image registration.
DistillMatch [9]	2025	VFM teacher (DINOv2/DINOv3) → knowledge-distilled lightweight student → modality information injection → V2I-GAN augmentation → matching	Uses VFM knowledge distillation to learn robust cross-modal features while retaining modality-specific information.	Primarily evaluated on visible/infrared-type multimodal data.

PROPOSED METHODOLOGY

Inference

- Georeferenced Image Input
- Correlation performed by Matcher
- Estimating Position Lat-Long
- Calculate error & repeatability

Fine-tuning Teacher

- **Inputs**
- **RGB-SAR Pair Dataset**
- **Preprocessing**
  (Common ground resolution, Normalisation, No-data Masking)
- **Matcher: MatchAnything-ELoFTR**
  (CNN backbone, self/cross attention ,coarse-to-fine matches)
- **Geometric Verification**
  (RANSAC[17] (similarity model), inlier-count accept/reject)
- **Position Fix**
  (Patch centre to geo-coordinates; error in pixels & metres)

Training Student

- **Inputs**
- **RGB-SAR Pair Dataset**
- **Preprocessing**
  (Common ground resolution, Normalisation, No-data Masking)
- **Student Training**
  (distillation loss + supervised loss; train & test on spatially separate regions)
- **Compact Student Matcher**
  (Replaces the matcher in teacher pipeline)

Edge Optimisation

- **Compress Teacher**
- **Optimise Student**
- **Benchmark on edge device**
  (Latency Parameters, Accuracy, NN mAP, Repeatability, Corner-loss Error to be compared with the teacher)

APPROACH FOR IMPLEMENTATION

Steps Implemented:

1. Dataset & Geospatial Preparation

- ICEYE SAR[10] + Google RGB imagery
- Common geographic coverage
- SAR resampled to RGB grid
- CRS/georeferencing verified
- SAR no-data masking + percentile normalization

2. Patch Generation

- Geographically co-located SAR–RGB patches
- Controlled patch size: 2048×2048 for MatchAnything experiments
- Smaller patches used during student experiments
- Training/holdout spatial split

3. Baseline Matcher MatchAnything–ELoFTR

- SAR–RGB cross-modal correspondence extraction
- Sliding RGB search windows
- Similarity/affine geometric verification using RANSAC [13]

4. Geolocation Evaluation

- Predicted patch centre → RGB pixel coordinates
- RGB pixel coordinates → latitude/longitude
- Geodesic localization error in metres
- Metrics: matches, inliers, pixel error, localization error

Steps Next to be Implemented:

5. RGB-SAR Pairs Dataset [11] for Student Model

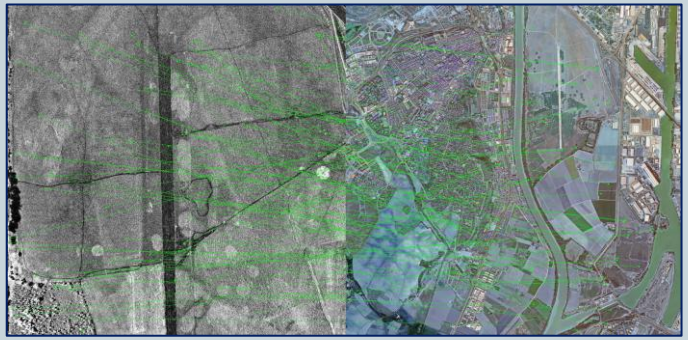
6. Distillation Training[12]

7. Edge Optimisation

PRELIMINARY RESULTS

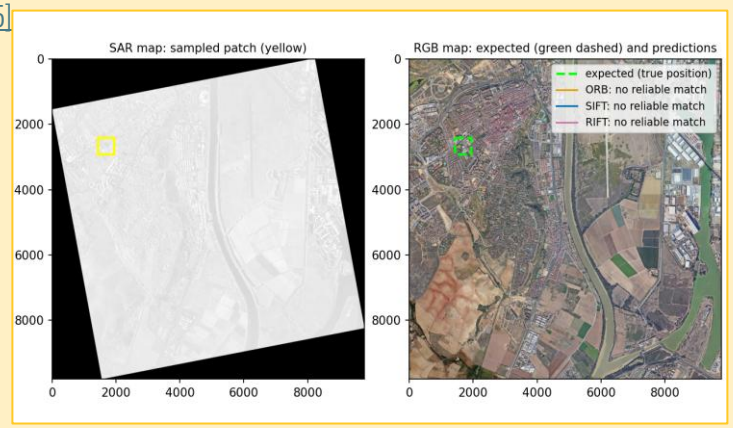
ELoFTR

- Irrelevant Scene Matching
  No reliable match found
- Reason: ELoFTR not inherently designed for cross modality

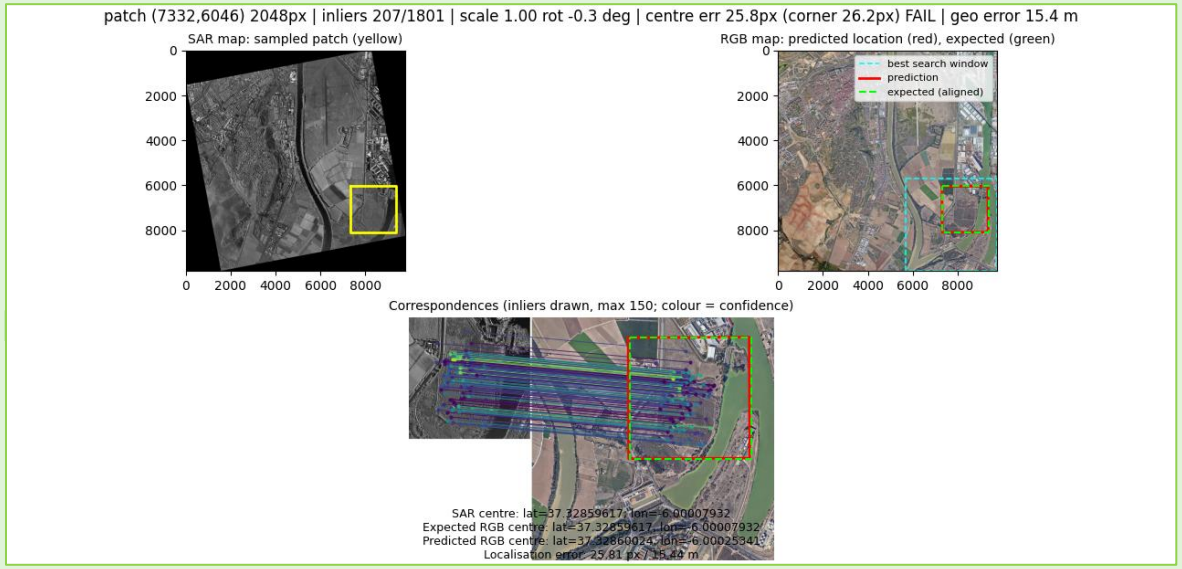


SIFT [13], ORB [14], RIFT [15]

- No reliable match found in any method
- Reason: handcrafted local features lack modality-invariant descriptors.



MatchAnything-ELoFTR



SAR centre: lat=37.32859617;lon =- 6.00007932
Expected RGB centre: lat=37.32859617.lon=6.00007932
Predicted RGB centre: lat=37.32860024. lon =- 6.00025341
Localisation error 25.81 px / 15.44 m
Threshold: 10m [16]

SUMMARY & CONCLUSION

Summary

- Cross-modal image matching is an important component for robust DSMAC operation in GPS-denied environments.
- The literature shows significant progress in learned and cross-modal matching, but **SAR–RGB matching remains challenging due to the large differences in imaging characteristics between the two modalities**.
- Modern methods such as MatchAnything demonstrate strong cross-modal matching capability, while DistillMatch shows the potential of **knowledge distillation for developing lightweight multimodal matching models**.
- Our preliminary implementation establishes a **SAR–RGB matching and geolocation evaluation pipeline**, with MatchAnything–ELoFTR achieving a localization error of **15.44 m** on the tested scene.
- These results motivate the development of a **dedicated lightweight SAR–RGB matcher** rather than directly relying on a general-purpose pretrained matcher.

Conclusion

- The proposed direction is to develop a **lightweight cross-modal student model**, trained using geospatially aligned SAR–RGB data and knowledge transferred from a strong teacher matcher.
- The framework will focus on learning **cross-modal correspondence and geometric information**, while retaining modality-specific characteristics important for SAR–RGB matching.
- The overall objective is to achieve a practical **accuracy–efficiency trade-off suitable for edge/onboard deployment**.

REFERENCES

[1] J. R. Carr and J. S. Sobek, “Digital Scene Matching Area Correlator (DSMAC),” in Proc. SPIE 0238, Image Processing for Missile Guidance, San Diego, CA, USA, Dec. 1980, pp. 36–41. doi: 10.1117/12.959130.

[2] G. B. Irani and J. P. Christ, "Image processing for Tomahawk scene matching," Johns Hopkins APL Technical Digest, vol. 15, no. 3, pp. 250–264, Jul.–Sep. 1994.

[3] S. Kashyap, J. Shivamurthy, and C. R. S. Kumar, "ML Based Solution for DSMAC System in Aerial Navigation," in ML Based Solution for DSMAC System in Aerial Navigation, Springer, 2026, ch. 23.

[4] J. Sun, Z. Shen, Y. Wang, H. Bao, and X. Zhou, "LoFTR: Detector-Free Local Feature Matching with Transformers," in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2021, pp. 8922–8931, doi: 10.1109/CVPR46437.2021.00882.

[5] Y. Wang, X. He, S. Peng, D. Tan, and X. Zhou, "Efficient LoFTR: Semi-Dense Local Feature Matching with Sparse-Like Speed," in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Seattle, WA, USA, 2024, pp. 2047, doi: 10.1109/CVPR52733.2024.02047.

[6] Ö. Tuzcuoğlu, A. Köksal, B. Sofu, S. Kalkan and A. A. Alatan, "XoFTR: Cross-modal Feature Matching Transformer," 2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), Seattle, WA, USA, 2024, pp. 4275-4286, doi: 10.1109/CVPRW63382.2024.00431. keywords: {Learning systems;Image matching;Pipelines;Lighting;Benchmark testing;Transformers;Image augmentation;feature matching;cross modality;transformer;infrared;thermal infrared},

[7] J. Chen, H. Xie, L. Zhang, J. Hu, H. Jiang, and G. Wang, "SAR and Optical Image Registration Based on Deep Learning with Co-Attention Matching Module," Remote Sensing, vol. 15, no. 15, p. 3879, Aug. 2023, doi: 10.3390/rs15153879.

[8] S. Li et al., "Matching Anything by Segmenting Anything," 2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Seattle, WA, USA, 2024, pp. 18963-18973, doi: 10.1109/CVPR52733.2024.01794. keywords: {Image segmentation;Adaptation models;Shape;Video sequences;Object segmentation;Benchmark testing;Data models},

[9] C. Wang, B. Zhao and Z. Liu, "DistillMatch: Revisiting Self-Knowledge Distillation in Semi-Supervised Medical Image Segmentation," 2024 IEEE International Conference on Bioinformatics and Biomedicine (BIBM), Lisbon, Portugal, 2024, pp. 3778-3783, doi: 10.1109/BIBM62325.2024.10821816. keywords: {Image segmentation;Perturbation methods;Image edge detection;Computational modeling;Benchmark testing;Data models;Computational efficiency;Knowledge transfer;Faces;Biomedical imaging;Semi-supervised segmentation;Self-knowledge distillation},

REFERENCES

[10] ICEYE, "ICEYE Synthetic Aperture Radar (SAR) Open Dataset," ICEYE Open Data Catalog via AWS, [Online]. Available: https://registry.opendata.aws/iceye-opendata. [Accessed: Oct. 6, 2026].

[11] xiang yu ming, Chen Jinyang, Hong Zhonghua, et al. OSDataset2.0: Optical-SAR Image Matching Dataset and Evaluation Benchmark[DS/OL]. V1. Science Data Bank, 2025[2026-10-07]. https://cstr.cn/31253.11.sciencedb.radars.00061. CSTR:31253.11.sciencedb.radars.00061.

[12] A. A. Neeralgi, R. R. Avarsekar, M. Bhajantri, R. Khatod, U. Kulkarni and S. M. Nadaf, "Knowledge Distillation Using Deep Learning Techniques: A Survey," 2024 IEEE Conference on Engineering Informatics (ICEI), Melbourne, Australia, 2024, pp. 1-10, doi: 10.1109/ICEI64305.2024.10912425. keywords: {Deep learning;Training;Adaptation models;Uncertainty;Computational modeling;Artificial neural networks;Natural language processing;Data models;Usability;Optimization},

[13] F. Guo, J. Yang, Y. Chen and B. Yao, "Research on image detection and matching based on SIFT features," 2018 3rd International Conference on Control and Robotics Engineering (ICCRE), Nagoya, Japan, 2018, pp. 130-134, doi: 10.1109/ICCRE.2018.8376448. keywords: {Feature extraction;Data models;Image recognition;Marine vehicles;Image registration;Image matching;Transforms;SIFT;Features points;Match;RANSAC},

[14] E. Rublee, V. Rabaud, K. Konolige, and G. Bradski, "ORB: An efficient alternative to SIFT or SURF," in 2011 International Conference on Computer Vision, Barcelona, Spain, 2011, pp. 2564–2571, doi: 10.1109/ICCV.2011.6126544.

[15] J. Li, Q. Hu, and M. Ai, "RIFT: Multi-modal image matching based on radiation-variation insensitive feature transform," IEEE Transactions on Image Processing, vol. 29, pp. 3296–3310, 2020, doi: 10.1109/TIP.2020.2992976.

[16] https://www.designation-systems.net/dusrm/m-109.html

[17] M. A. Fischler and R. C. Bolles, "Random sample consensus: A paradigm for model fitting with applications to image analysis and automated cartography," Communications of the ACM, vol. 24, no. 6, pp. 381-395, June 1981, doi: 10.1145/358669.358692.